

Chung-Kuan Cheng

professor

基本信息

姓名: Chung-Kuan Cheng

所属院校: 加州大学-圣地亚哥

邮箱: ckcheng@ucsd.edu

地区: United States

年龄范围: 50岁以上

企业经历: 有

学科领域

电子与计算机工程

电气工程及其自动化

集成电路设计与集成系统

微电子科学与工程

智能电网信息工程

研究领域

集成电路设计与制造

电力系统仿真与优化

计算机体系结构

电路与系统设计

EDA工具与电路验证

数字集成电路设计

个人简介

Chung-Kuan Cheng is a Distinguished Professor at the Department of Computer Science and Engineering and an Adjunct Professor at the Department of Electrical and Computer Engineering at the University of California, San Diego. He received his Ph.D. from the University of California, Berkeley in 1984. Prior to his academic career, he served as a Senior Engineer and Consultant at various System, Design, and Electronic Design Automation Companies. His research focuses on multiple areas within computer science and engineering. He can be reached at ckcheng at ucsd dot edu.

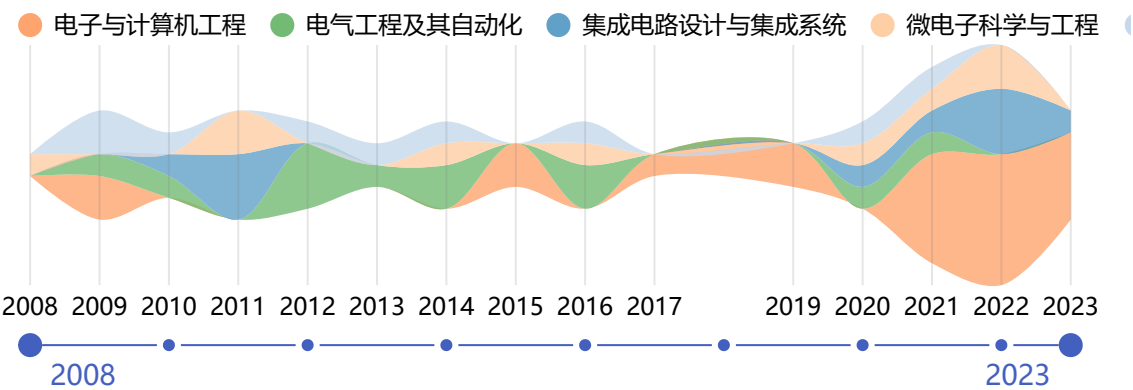
工作经历

教育经历

University of California Berkeley - Ph.D.

1980-09-01 - 1984-06-30

近年发表的论文



Using EEG signals to assess workload during memory retrieval in a real-world scenario

JOURNAL OF NEURAL ENGINEERING

Chiang, Kuan-Jung; Dong, Steven; Cheng, Chung-Kuan; Jung, Tzyy-Ping

被引用次数: 2

2023

Invited Paper: The Scope and Challenges of Scaling in Advanced Technologies

2023 ACM/IEEE SYSTEM LEVEL INTERCONNECT PATHFINDING WORKSHOP, SLIP 2023

Cheng, Chung-Kuan; Lin, Bill; Kang, Byeonggon; Wang, Yucheng

被引用次数: 0

2023

Gear-Ratio-Aware Standard Cell Layout Framework for DTCO Exploration

2023 ACM/IEEE SYSTEM LEVEL INTERCONNECT PATHFINDING WORKSHOP, SLIP 2023

Cheng, Chung-Kuan; Kahng, Andrew B.; Lin, Bill; Wang, Yucheng; Yoon, Dooseok

被引用次数: 0

2023

DAGSizer: A Directed Graph Convolutional Network Approach to Discrete Gate Sizing of VLSI Graphs

ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS

Cheng, Chung-Kuan; Holtz, Chester; Kahng, Andrew B.; Lin, Bill; Mallappa, Uday

被引用次数: 2

2023

Placement Initialization via Sequential Subspace Optimization with Sphere Constraints

PROCEEDINGS OF THE 2023 INTERNATIONAL SYMPOSIUM ON PHYSICAL DESIGN, ISPD 2023

Chen, Pengwen; Cheng, Chung-Kuan; Chern, Albert; Holtz, Chester; Li, Aoxi; Wang, Yucheng

被引用次数: 0

2023

Assessment of Reinforcement Learning for Macro Placement

PROCEEDINGS OF THE 2023 INTERNATIONAL SYMPOSIUM ON PHYSICAL DESIGN, ISPD 2023 Cheng, Chung-Kuan; Kahng, Andrew B.; Kundu, Sayak; Wang, Yucheng; Wang, Zhiang 被引用次数: 13	2023
Machine Learning Prediction for Design and System Technology Co-Optimization Sensitivity Analysis IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS Cheng, Chung-Kuan; Ho, Chia-Tung; Holtz, Chester; Lee, Daeyeal; Lin, Bill 被引用次数: 5	2022
JARVA: Joint Application-Aware Oblivious Routing and Static Virtual Channel Allocation IEEE DESIGN & TEST Mallappa, Uday; Cheng, Chung-Kuan; Lin, Bill 被引用次数: 1	2022
Joint Application-Aware Oblivious Routing and Static Virtual Channel Allocation IEEE EMBEDDED SYSTEMS LETTERS Mallappa, Uday; Cheng, Chung-Kuan; Lin, Bill 被引用次数: 0	2022
Monolithic 3D Semiconductor Footprint Scaling Exploration Based on VFET Standard Cell Layout Methodology, Design Flow, and EDA Platform IEEE ACCESS Cheng, Chung-Kuan; Ho, Chia-Tung; Lee, Daeyeal; Lin, Bill 被引用次数: 3	2022
SMT-Based Contention-Free Task Mapping and Scheduling on 2D/3D SMART NoC with Mixed Dimension-Order Routing ACM TRANSACTIONS ON ARCHITECTURE AND CODE OPTIMIZATION Lee, Daeyeal; Lin, Bill; Cheng, Chung-Kuan 被引用次数: 3	2022
Net Separation-Oriented Printed Circuit Board Placement via Margin Maximization 27TH ASIA AND SOUTH PACIFIC DESIGN AUTOMATION CONFERENCE, ASP-DAC 2022 Cheng, Chung-Kuan; Ho, Chia-Tung; Holtz, Chester 被引用次数: 3	2022
Late Breaking Results: Placement Initialization via a Projected Eigenvector Algorithm PROCEEDINGS OF THE 59TH ACM/IEEE DESIGN AUTOMATION CONFERENCE, DAC 2022 Chen, Pengwen; Cheng, Chung-Kuan; Chern, Albert; Holtz, Chester; Li, Aoxi; Wang, Yucheng 被引用次数: 2	2022
Placement initialization via a projected eigenvector algorithm: late breaking results Design Automation Conference Pengwen Chen; Chung-Kuan Cheng; Albert Chern; Chester Holtz; Aoxi Li; Yucheng Wang 被引用次数: 0	2022

PROBE2.0: A Systematic Framework for Routability Assessment From Technology to Design in Advanced Nodes

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Cheng, Chung-Kuan; Kahng, Andrew B.; Kim, Hayoung; Kim, Minsoo; Lee, Daeyeal; Park, Dongwon; Woo, Mingyu

被引用次数: 15

2022

Design and System Technology Co-Optimization Sensitivity Prediction for VLSI Technology Development using Machine Learning

2021 ACM/IEEE INTERNATIONAL WORKSHOP ON SYSTEM-LEVEL INTERCONNECT PATHFINDING (SLIP 2021)

Cheng, Chung-Kuan; Ho, Chia-Tung; Holtz, Chester; Lin, Bill

被引用次数: 15

2021

Complementary-FET (CFET) Standard Cell Synthesis Framework for Design and System Technology Co-Optimization Using SMT

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Cheng, Chung-Kuan; Ho, Chia-Tung; Lee, Daeyeal; Lin, Bill; Park, Dongwon

被引用次数: 26

2021

ARNOLDI ALGORITHMS WITH STRUCTURED ORTHOGONALIZATION

SIAM JOURNAL ON NUMERICAL ANALYSIS

Chen, Pengwen; Cheng, Chung-Kuan; Wang, Xinyuan

被引用次数: 0

2021

Epidermal Graphene Sensors and Machine Learning for Estimating Swallowed Volume

ACS APPLIED NANO MATERIALS

Polat, Beril; Becerra, Laura L.; Hsu, Po-Ya; Kaipu, Vineel; Mercier, Patrick P.; Cheng, Chung-Kuan; Lipomi, Darren J.

被引用次数: 18

2021

SP&R: SMT-Based Simultaneous Place-and-Route for Standard Cell Synthesis of Advanced Nodes

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Lee, Daeyeal; Park, Dongwon; Ho, Chia-Tung; Kang, Ilgweon; Kim, Hayoung; Gao, Sicun; Lin, Bill; Cheng, Chung-Kuan

被引用次数: 12

2021

SMT-Based Contention-Free Task Mapping and Scheduling on SMART NoC

IEEE EMBEDDED SYSTEMS LETTERS

Lee, Daeyeal; Lin, Bill; Cheng, Chung-Kuan

被引用次数: 6

2021

SAT-Based On-Track Bus Routing

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Zhang, He-Teng; Fujita, Masahiro; Cheng, Chung-Kuan; Jiang, Jie-Hong R.

CoRe-ECO: Concurrent Refinement of Detailed Place-and-Route for an Efficient ECO Automation

2021 IEEE 39TH INTERNATIONAL CONFERENCE ON COMPUTER DESIGN (ICCD 2021)

Cheng, Chung-Kuan; Kahng, Andrew B.; Kang, Ilgweon; Kim, Minsoo; Lee, Daeyeal; Lin, Bill; Park, Dongwon; Woo, Mingyu

被引用次数: 4

2021

Grid-Based Framework for Routability Analysis and Diagnosis With Conditional Design Rules

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Park, Dongwon; Lee, Daeyeal; Kang, Ilgweon; Holtz, Chester; Gao, Sicun; Lin, Bill; Cheng, Chung-Kuan

被引用次数: 15

2020

Stability and Convergency Exploration of Matrix Exponential Integration on Power Delivery Network Transient Simulation

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Wang, Xinyuan; Chen, Pengwen; Cheng, Chung-Kuan

被引用次数: 2

2020

Three-dimensional Floorplan Representations by Using Corner Links and Partial Order

ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS

Kang, Ilgweon; Qiao, Fang; Park, Dongwon; Kane, Daniel; Young, Evangeline Fung Yu; Cheng, Chung-Kuan; Graham, Ronald

被引用次数: 1

2019

RePlAcE: Advancing Solution Quality and Routability Validation in Global Placement

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Cheng, Chung-Kuan; Kahng, Andrew B.; Kang, Ilgweon; Wang, Lutong

被引用次数: 107

2019

An Online Brain-Computer Interface Based on SSVEPs Measured From Non-Hair-Bearing Areas

IEEE TRANSACTIONS ON NEURAL SYSTEMS AND REHABILITATION ENGINEERING

Wang, Yu-Te; Nakanishi, Masaki; Wang, Yijun; Wei, Chun-Shu; Cheng, Chung-Kuan; Jung, Tzyy-Ping

被引用次数: 47

2017

A fast time-domain EM-TCAD coupled simulation framework via matrix exponential with stiffness reduction

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Chen, Quan; Schoenmaker, Wim; Weng, Shih-Hung; Cheng, Chung-Kuan; Chen, Guan-Hua; Jiang, Li-Jun; Wong, Ngai

被引用次数: 1

2016

Simulation Algorithms With Exponential Integration for Time-Domain Analysis of Large-Scale Power Delivery Networks

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Zhuang, Hao; Yu, Wenjian; Weng, Shih-Hung; Kang, Ilgweon; Lin, Jeng-Hau; Zhang, Xiang; Coutts, Ryan; Cheng, Chung-Kuan

被引用次数: 15

2016

An Efficient Transient Electro-Thermal Simulation Framework for Power Integrated Circuits

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Mei, Qinggao; Schoenmaker, Wim; Weng, Shih-Hung; Zhuang, Hao; Cheng, Chung-Kuan; Chen, Quan

被引用次数: 10

2016

From Circuit Theory, Simulation to SPICEDiego: A Matrix Exponential Approach for Time-Domain Analysis of Large-Scale Circuits

IEEE CIRCUITS AND SYSTEMS MAGAZINE

Zhuang, Hao; Wang, Xinyuan; Chen, Quan; Chen, Pengwen; Cheng, Chung-Kuan

被引用次数: 8

2016

ePlace-MS: Electrostatics-Based Placement for Mixed-Size Circuits

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Lu, Jingwei; Zhuang, Hao; Chen, Pengwen; Chang, Hongliang; Chang, Chin-Chih; Wong, Yiu-Chung; Sha, Lu; Huang, Dennis; Luo, Yufeng; Teng, Chin-Chi; Cheng, Chung-Kuan

被引用次数: 74

2015

ePlace: Electrostatics-Based Placement Using Fast Fourier Transform and Nesterov's Method

ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS

Lu, Jingwei; Chen, Pengwen; Chang, Chin-Chih; Sha, Lu; Huang, Dennis Jen-Hsin; Teng, Chin-Chi; Cheng, Chung-Kuan

被引用次数: 94

2015

Energy Efficiency Optimization Through Codesign of the Transmitter and Receiver in High-Speed On-Chip Interconnects

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Weng, Shih-Hung; Zhang, Yulei; Buckwalter, James F.; Cheng, Chung-Kuan

被引用次数: 13

2014

Worst Case Noise Prediction With Nonzero Current Transition Times for Power Grid Planning

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Hu, Xiang; Du, Peng; Weng, Shih-Hung; Cheng, Chung-Kuan

被引用次数: 0

2014

Developing an EEG-based on-line closed-loop lapse detection and mitigation system

FRONTIERS IN NEUROSCIENCE

Wang, Yu-Te; Huang, Kuan-Chih; Wei, Chun-Shu; Huang, Teng-Yi; Ke, Li-Wei; Lin, Chin-Teng; Cheng, Chung-Kuan; Jung, Tzyy-Ping

被引用次数: 26

2014

Ratio of the Worst Case Noise and the Impedance of Power Distribution Network

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Zhang, Xiang; Liu, Yang; Hu, Xiang; Cheng, Chung-Kuan

被引用次数: 2

2014

Modeling and Analysis of Power Distribution Networks in 3-D ICs

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Hu, Xiang; Du, Peng; Buckwalter, James F.; Cheng, Chung-Kuan

被引用次数: 12

2013

Novel Differential-Mode Equalizer With Broadband Common-Mode Filtering for Gb/s Differential-Signal Transmission

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Cheng, Yu-Jen; Chuang, Hao-Hsiang; Cheng, Chung-Kuan; Wu, Tzong-Lin

被引用次数: 22

2013

A Realistic Early-Stage Power Grid Verification Algorithm Based on Hierarchical Constraints

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Wang, Yuanzhe; Hu, Xiang; Cheng, Chung-Kuan; Pang, Grantham K. H.; Wong, Ngai

被引用次数: 22

2012

Time-Domain Analysis of Large-Scale Circuits by Matrix Exponential Method With Adaptive Control

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Weng, Shih-Hung; Chen, Quan; Cheng, Chung-Kuan

被引用次数: 29

2012

A Practical Regularization Technique for Modified Nodal Analysis in Large-Scale Time-Domain Circuit Simulation

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Chen, Quan; Weng, Shih-Hung; Cheng, Chung-Kuan

被引用次数: 28

2012

Bus Matrix Synthesis Based on Steiner Graphs for Power Efficient System-on-Chip Communications

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Wang, Renshen; Zhang, Yulei; Chou, Nan-Chi; Young, Evangeline F. Y.; Cheng, Chung-Kuan; Graham, Ronald

被引用次数: 2

2011

Analysis and Optimization of Low-Power Passive Equalizers for CPU-Memory Links

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Zhang, Ling; Yu, Wenjian; Zhang, Yulei; Wang, Renshen; Deutsch, Alina; Katopis, George A.; Dreps, Daniel M.; Buckwalter, James; Kuh, Ernest S.; Cheng, Chung-Kuan

被引用次数: 2

2011

On-Chip Interconnect Analysis of Performance and Energy Metrics Under Different Design Goals

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Zhang, Ling; Zhang, Yulei; Chen, Hongyu; Yao, Bo; Hamilton, Kevin; Cheng, Chung-Kuan

被引用次数: 8

2011

Prediction and Comparison of High-Performance On-Chip Global Interconnection

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Zhang, Yulei; Hu, Xiang; Deutsch, Alina; Engin, A. Ege; Buckwalter, James F.; Cheng, Chung-Kuan

被引用次数: 16

2011

Efficient Power Network Analysis with Modeling of Inductive Effects

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Zeng, Shan; Yu, Wenjian; Hong, Xianlong; Cheng, Chung-Kuan

被引用次数: 1

2010

Complexity of 3-D Floorplans by Analysis of Graph Cuboidal Dual Hardness

ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS

Wang, Renshen; Young, Evangeline; Cheng, Chung-Kuan

被引用次数: 3

2010

Accurate Eye Diagram Prediction Based on Step Response and Its Application to Low-Power Equalizer Design

IEICE TRANSACTIONS ON ELECTRONICS

Yu, Wenjian; Shi, Rui; Cheng, Chung-Kuan

被引用次数: 13

2009

Efficient Power Network Analysis Considering Multidomain Clock Gating

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Zhang, Wanping; Yu, Wenjian; Hu, Xiang; Zhang, Ling; Shi, Rui; Peng, He; Zhu, Zhi; Chua-Eoan, Lew; Murgai, Rajeev; Shibuya, Toshiyuki; Ito, Noriyuki; Cheng, Chung-Kuan

被引用次数: 9

2009

Energy and Switch Area Optimizations for FPGA Global Routing Architectures

ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS

Zhu, Yi; Hu, Yuanfang; Taylor, Michael B.; Cheng, Chung-Kuan

被引用次数: 6

2009

Efficient Partial Reluctance Extraction for Large-Scale Regular Power Grid Structures

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Zeng, Shan; Yu, Wenjian; Shi, Jin; Hong, Xianlong; Cheng, Chung-Kuan

被引用次数: 4

2009

Enhancing Learning Effectiveness in Digital Design Courses Through the Use of Programmable Logic Boards

IEEE TRANSACTIONS ON EDUCATION

Zhu, Yi; Weng, Thomas; Cheng, Chung-Kuan

被引用次数: 302009

Analyzing High-Density ECG Signals Using ICA

IEEE TRANSACTIONS ON BIOMEDICAL ENGINEERING

Zhu, Yi; Shayan, Amirali; Zhang, Wanping; Chen, Tong Lee; Jung, Tzyy-Ping; Duann, Jeng-Ren; Makeig, Scott; Cheng, Chung-Kuan

被引用次数: 132008

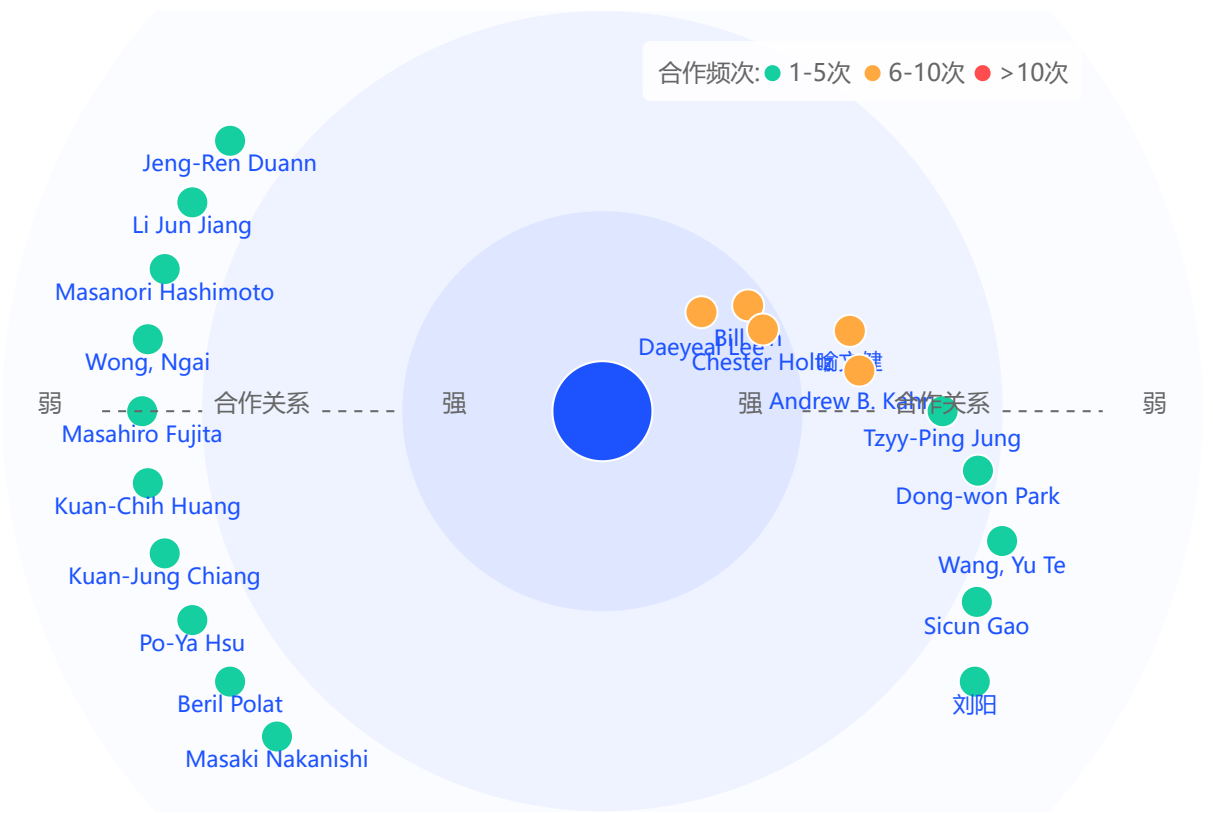
Analytical Eye-Diagram Model for On-Chip Distortionless Transmission Lines and Its Application to Design Space Exploration

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Hashimoto, Masanori; Siriporn, Jangsombatsiri; Tsuchiya, Akira; Zhu, Haikun; Cheng, Chung-Kuan

被引用次数: 52008

学术合作



Daeyeal Lee

加州大学-圣地亚哥

合作次数: 9

Bill Lin 加州大学-圣地亚哥	合作次数: 8
Chester Holtz 加州大学-圣地亚哥	合作次数: 8
喻文健 清华大学	合作次数: 6
Andrew B. Kahng 加州大学-圣地亚哥	合作次数: 6

还有 16 个合作者